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**TABLE II**  
**Operational Ranges of IEEE 802.15.4**

| Frequency band | Geography | Data rate | Available channels |
|----------------|-----------|-----------|--------------------|
| 868 MHz        | Europe    | 20 kbps   | 1                  |
| 915 MHz        | Americas  | 40 kbps   | 10                 |
| 2.4 GHz        | Worldwide | 250 kbps  | 16                 |

ally needed. Sometimes equipment are moved or stored, and can be difficult to find when needed—especially in an emergency situation. This increases the costs for hospitals that are unnecessary when an asset-tracking system is used. Equipment can be tracked through triangulation methods to determine their exact location.

### IEEE 802.15.4: Ideal for smaller apps

The other wireless networking technology that should be considered for medical applications is IEEE 802.15.4. This is often used in reference to the ZigBee standard, but the ZigBee standard is only a protocol that uses IEEE 802.15.4 as the physical medium.

IEEE 802.15.4 specifies three frequency bands designated for use in industrial, scientific and medical (ISM) applications: 868 MHz, 915 MHz and 2.4 GHz. Table II shows the data rate and number of channels for each frequency. The geographies stated are the major regions that allow use of the frequency band. However, there are exceptions.

**Protocols.** There are several protocols that use IEEE 802.15.4. Among them, ZigBee is probably the best known protocol, but it's still in infancy in terms of widespread deployment. Other protocols do exist, such as the simple media access controller (SMAC) protocol from Freescale Semiconductor and the MiWi protocol from Microchip Technology.

The ZigBee protocol mandates interoperability, which is one of its best features. Under the ZigBee protocol, different devices from different manufacturers can communicate with each other and interact in a star or mesh network.

ZigBee protocol networks can consist of over 65,000 devices and allow the dynamic formation and unlimited routing of messages through the network. All of this functionality does not come for free. Typically, ZigBee protocol stacks for network coordinators range in size from 64 kB to 96 kB and require a substantial amount of RAM and non-volatile memory to store the routing information. End devices are typically in the range of 32 kB to 64 kB.

Applications that don't require the interoperability or large network size offered by the ZigBee protocol can use the SMAC or MiWi protocols. SMAC provides a stack size of 2 kB and limits the network topology to point-to-point or star. The MiWi protocol provides more flexibility, with the capability of supporting over 1000 nodes and point-to-point, star and limited mesh topologies. The size of the MiWi protocol stack is 20 kB for a coordinator and 4 kB for an end device.

**Applications.** Devices using the IEEE 802.15.4 protocol are ideal for smaller medical applications like blood-pressure monitors, weighing scales, glucometers, spirometers, thermometers and emergency signals. With costs approaching the sub-\$2 range for the transceiver and options for small software stacks that enable the use of very low-cost microcontrollers, adding wireless capabilities to these products results in a price appropriate for the residential or nursing-home markets.

**Most hospitals already have a WiFi network deployed, and equipment such as ECG monitors, pulse oximeters, blood pressure monitors and infusion pumps can now implement an IEEE 802.11 transceiver cost-effectively**